Mehrshad Saadatinia

Graduate Student at University of Southern California

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Education

University of Southern California — **Jan. 2024 – Present**
MS in Computer Science — *Los Angeles, California, USA*

- GPA over last two semesters: **3.92/4.0**

Shahid Beheshti University — **Sep. 2018 – Feb. 2023**
BSc in Computer Engineering — *Tehran, Iran*

- Cumulative GPA: 17.40/20 (3.7/4) | **Best B.Sc thesis**

Research Interests

Deep Learning | Representation Learning | Bioinformatics | Large Language Models

Publications

- "An Explainable Deep Learning-Based Method For Schizophrenia Diagnosis Using Generative Data-Augmentation" **M Saadatinia** and A Salimi-Badr (2024). *IEEE Access* [link]
- "Multi-organ metabolome biological age implicates cardiometabolic conditions and mortality risk" F Anagnostakis, S Ko, **M Saadatinia**, J Wang, C Davatzikos, J Wen (2025). *Nature Communication* [link]
- "EvoGLAD: Evolving Graph-Level Representations for Anomaly Detection in Metagenomic Communities" **M Saadatinia**, Willie Neiswanger (2025). *AAAI-2026 (under review)*
- "Enhancing Multi-Modal Video Sentiment Classification Through Semi-Supervised Clustering" **M Saadatinia**, M Ahmadi, A Abdollahi (2025). *Arxiv preprint* [link]

Experience

Dr. Neiswanger's Lab at USC — **Jan. 2025 – Present**
Research Intern, Supervisor: ***Dr. Willie Neiswanger*** — *University of Southern California*

- [Project] Representing DNA/RNA Communities using Graph Neural Networks (GNNs) and a Metagenomic Foundation Model (METAGENE-1)
- Paper submitted to **AAAI-2026** main-track

Laboratory of AI & Biomedical Science (LABS) — **May. 2024 – Present**
Research Intern, Supervisor: ***Dr. Junhao Wen*** — *Columbia University*

- [Project] Developing a novel method to detect disease subtypes with imaging and genetic signatures via weakly-supervised deep clustering and Graph representations of MRI features, under the supervision of
- Paper under preparation for **ICLR-2026** submission

Center for Advanced Research Computing (CARC) — **April. 2025 – Present**
Student Worker, Supervisor: ***Dr. Iman Rahbari*** — *University of Southern California*

- [Project] Benchmarking, evaluating and improving Large Language Models for Deep Research
- [Project] Retrieval Augmented Generation (RAG) from CARC videos to improve CARC-LLM responses

Natural Language Processing Lab — **Sep. 2022 – Jan. 2023**
Undergraduate Research Assistant, Supervisor: ***Dr. Mehrnoush Shamsfard*** — *Shahid Beheshti University*

- [Project] Developed a Transformer-based model for informal-to-formal style transfer in Persian texts

Relevant Graduate Coursework

Deep Learning and Applications (A) | Applied Natural Language Processing (A)
Machine Learning (A-) | Advanced Computer Vision (A)

Skills

Programming Languages: Python, C/C++ (and OpenGL), R, Java, JavaScript, MATLAB
ML & DL: TensorFlow (and Keras), PyTorch, Scikit-Learn, LangChain
Technologies: Git, Linux, Docker, Google Cloud, Databases (SQL and MongoDB), Vector DBs, REST
Other Skills: Mathematics, Statistics, Generative Models, RAG

Teaching Experience

Head-TA, Artificial Intelligence & Expert Systems (Undergrad), Shahid Beheshti University, 3 Semesters
Head-TA, Signals and Systems (Undergrad), Shahid Beheshti University, 3 Semesters (2021–2023)
Head-TA, Advanced Programming (Undergrad), Shahid Beheshti University, 4 Semesters (2019–2021)
TA, Compiler Design (Undergrad), Shahid Beheshti University, 1 Semester (Fall 2020)

Selected Projects

Deep Learning-Based Schizophrenia Diagnosis | Code | App code

- Developed a deep CNN-based system for schizophrenia diagnosis, integrating a VAE-based data augmentation model improving diagnostic accuracy by 3%, reaching 99%

LoRA Fine-tuned Stable Diffusion for High-Fidelity Human Face Generation | Code

- Fine-tuned Stable Diffusion with LoRA for high-fidelity human face synthesis, achieving significant improvements in realism and detail over the base model
- Preparing model for public release on Hugging Face

Interpretable Multi Modal Depression Detection | Code

- Automatic Depression Detection Using An Interpretable Audio-textual Multi-modal Transformer-based Model
- Improved state-of-the-art accuracy by 2.0% on DAIC-WoZ dataset

Deep Embedded Clustering | Code

- A re-implementation of the Deep Embedded Clustering paper in PyTorch, reproducing original results

Honors & Awards

The **best B.Sc thesis award** in CSE department | certificate

Ranked within the top **0.2%** among 300,000 participants (2018 nationwide university entrance exam)

Certifications and Workshops

Machine Learning Course | Certificate — **February 2022**
Instructed by: Andrew Ng — *Stanford Online | Coursera*
Deep Learning Specialization | 5 Courses | Certificate — **September 2023**
Instructed by: Andrew Ng — *DeepLearning.AI | Coursera*
Build Basic Generative Adversarial Networks | Certificate — **April 2023**
Instructed by: Sharon Zhou — *DeepLearning.AI | Coursera*
NLP lab Summer Workshop | Certificate — **July 2022**
Instructed by: Dr. Mehrnoush Shamsfard — *SBU NLP lab*

References

Dr. Willie Neiswanger | Assistant Professor
Email: neiswang@usc.edu
Dr. Junhao Wen | Assistant Professor
Email: jw4745@cumc.columbia.edu
Dr. Armin Salimi-Badr | Assistant Professor
Email: a_salimibadr@sbu.ac.ir
Dr. Iman Rahbari | Computational Scientist
Email: irahbari@usc.edu